

QUESTION PAPER

Name of the Examination: Fall 2023-24 Semester – CAT-1

Course Code: CSE4011

Course Title: Internet of Things

Set number: 2

Date of Exam: 12/09/2023 (AN) (C2)

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

- Q1.** What are the fundamental distinctions between IoT networks and M2M networks in terms of their architecture, communication protocols, and applications? How do these differences impact the deployment, management, and scalability of these two types of networks in various use cases? **(10M)**
- Q2.** What is Software-Defined Networking (SDN), and how does it differ from traditional network architectures? Could you provide a high-level overview of the core concepts and advantages that SDN brings to modern networking? **(10M)**
- Q3.** What are some of the primary challenges and issues associated with medium access control in wireless communication systems? How do factors like contention, interference, and channel access methods impact the performance and reliability of wireless networks, and what are some strategies to mitigate these issues? **(10M)**
- Q4.** How do sensing, actuation, and control mechanisms function within IoT networks, and what are their roles in enabling smart and responsive applications? Can you provide examples of how these elements work together to create value in various IoT use cases, such as smart homes, industrial automation, or healthcare monitoring? **(10M)**
- Q5.** Discuss the advantages and trade-offs of popular IoT communication protocols like MQTT, CoAP, and HTTP/HTTPS, and how their suitability varies across different IoT applications and use cases. **(10M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	2	2	1,2,3,4	-	-	10
Q2	2	2	1,2,3,4	-	-	10
Q3	2	2	1,2,3,4	-	-	10
Q4	1	1	1,2,3,4,5,6	-	-	10
Q5	1	1	1,2,3,4,5,6	-	-	10